

The Tensegrity Knowledge Graph: A Holonic Architecture for Scientific Knowledge Representation

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Abstract

We present the Tensegrity Knowledge Graph (GCT), a formally grounded architecture for the dynamic representation of scientific research knowledge. The GCT synthesizes seven classical philosophical traditions — Fuller's IVM cuboctahedron geometry (12-slot semantic schema), Koestler's holonic architecture (5 epistemological scales), Aristotle's four causes (edge typing), Whitehead's process philosophy (temporal epistemic decay), Wittgenstein's family resemblance (similarity metric), Porphyry's tree (taxonomic inference), and Boethius's logical maxims (dual validation–inference engine) — into a unified, domain-agnostic framework. Applied to a corpus of 1,660 knowledge nodes and 2,120 typed edges spanning 8 node types and 12 causal-slot categories, the GCT: (i) identified that 84.0% of structural nodes are documented with fewer than 2 of 4 Aristotelian causes; (ii) generated 335 formally sound inference proposals via the Boethian maxim engine with 0 logical violations; (iii) achieved 100% Wittgenstein family-resemblance coverage across all four artifact types within the family-eligible subset (67/67 experiments, 35/35 concepts, 18/18 papers, 10/10 paper_sections); and (iv) algebraically identified 26 god nodes ($VE \geq 0.5$) as the most evidentially central knowledge constructs. We demonstrate domain independence through an isomorphic instantiation in biomedical research management, and formally prove soundness of the maxim system with respect to holonic scale constraints.

Keywords: knowledge representation, formal ontology, holonic architecture, Aristotelian causation, Wittgenstein family resemblance, temporal knowledge graphs, Boethian maxims, research knowledge management

1. Introduction

Scientific research is fundamentally an activity of artifact production: hypotheses are proposed, experiments are designed to test them, results are obtained that either support or undermine the hypotheses, and claims are extracted from those results to populate the arguments of papers. Each of these artifacts stands in complex relations to the others — causal, taxonomic, evidential, and temporal — and the coherence of a research programme depends on those relations being navigable. Existing knowledge management solutions — wikis, spreadsheets, reference managers, property-graph databases — represent such artifacts as flat collections of properties and links without typed relational semantics. They cannot answer questions of the form: which experiments causally ground this theoretical claim, and are any of them in logical contradiction with each other? What is the formal specification — the abstract design — that this experimental protocol instantiates? Which nodes in the network have lost evidential support because the experiments that sustained them were conducted two years ago and have not been replicated? These questions are not operationally complex; they are structurally blocked. The tools lack the ontological vocabulary to represent causal structure, holonic scale, and temporal credence simultaneously within a single, query-able schema. The coherence problem is therefore not a matter of data volume or search quality but of representational poverty at the level of relational semantics.

The knowledge representation community has produced powerful foundational ontologies — BFO, DOLCE, GFO — that supply taxonomic and mereological rigour sufficient for many representational tasks. These frameworks operate, however, on static extensional fact bases and do not distinguish among Aristotle's four types of causation — formal, material, efficient, and final — which carry fundamentally different inferential licenses. Knowing that experiment E efficiently caused result R warrants an attribution claim; knowing that hypothesis H formally specifies protocol P warrants a design-compliance claim; conflating the two destroys the inferential structure the distinction is meant to carry. The provenance community, through PROV-O and related standards, tracks causal history but conflates efficient causation with material dependence and provides no mechanism for temporal credence decay — no way to represent that a node's evidential support has diminished as the experiments grounding it age without replication. Vector stores and embedding-based retrieval systems capture semantic similarity across documents but destroy directional causal structure: they cannot distinguish "A caused B" from "B caused A," and they provide no logical inference layer. Bai et al. (2024), in the closest work to the present paper, construct a chemistry knowledge graph using PROV-DM with domain-specific typing, but their schema is domain-locked, provides no holonic scale hierarchy, and applies no formal inference engine derived from the graph's own structural geometry. The result is that no existing system simultaneously represents causal type, holonic scale, temporal credence, and logical inference over a unified and formally grounded schema.

We present the Tensegrity Knowledge Graph (GCT), an architecture that addresses this gap by synthesising seven classical philosophical traditions into a single operationalisable system. Fuller's cuboctahedral geometry — the Isotropic Vector Matrix — provides a non-arbitrary 12-slot semantic schema Σ in which every slot corresponds to a geometrically distinct relational direction, ensuring that no causal dimension is privileged over any other in the schema's design. Koestler's holon concept supplies a five-level epistemological scale hierarchy ($s_{\blacksquare}$ through $s_{\blacksquare}$, from byte-level tokens to discourse-level arguments) within which every node is assigned a definite scale and cross-scale containment is represented by typed CONTAINER edges. Aristotle's four causes provide the edge-type vocabulary: formal causation is encoded as SPECIFICATION edges, material causation as SUBSTRATE edges, efficient causation as CAUSE and PREDECESSOR edges, and final causation as EFFECT and CONTAINER edges — each type licensing a distinct class of inference. Whitehead's process philosophy grounds the temporal decay operator $VE_decayed(n, t) = VE_harmonic(n) \times 2^{-(t - t_evidence) / t_halflife}$, which continuously discounts the

evidential value of nodes as their supporting experiments age without corroboration. Wittgenstein's family resemblance semantics is operationalised as a dual similarity metric: cosine similarity over co-occurrence vectors for concept nodes, and Jaccard similarity over neighbour sets for experimental artifact nodes, replacing the classical requirement for necessary-and-sufficient conditions with a use-constituted notion of conceptual proximity. The Porphyrian tree provides the taxonomic backbone from which CONTAINER edges inherit their transitivity and containment-closure properties. Boethius's topical maxims — originally conceived as the loci of valid dialectical inference — are formalised as twelve inference rules, one per schema slot, that generate proposals over the live graph and detect logical violations without requiring an external reasoner. We define the system formally as a typed directed multigraph $G = (N, E, \tau_N, \tau_E, \sigma, \lambda)$ equipped with the 12-slot schema Σ , the five-level holonic scale hierarchy, and the Boethian maxim engine. We prove soundness of the maxim system with respect to the holonic scale constraints, establish Aristotelian completeness as a computable property measurable on any node, and demonstrate that the convergence of these traditions is not architecturally forced but follows from the shared structural problem they each address: how to represent intentional artifacts — things made for reasons, by agents, from materials, toward purposes — in a way that supports inference and supports temporal revision.

We demonstrate the architecture on a corpus of 1,660 nodes and 2,120 typed edges drawn from an active doctoral research programme in neural architecture research. The maxim engine generated 335 inference proposals and detected zero logical violations, confirming both the productivity and consistency of the inference layer. A finding of particular theoretical interest is that 84.0% of structural nodes have fewer than two of the four Aristotelian causes documented, providing a direct and computable measure of the formal incompleteness latent in the research record and unavailable to any flat representational scheme. Wittgenstein family-resemblance coverage reaches 100% across all four artifact types within the family-eligible subset (67/67 experiments, 35/35 concepts, 18/18 papers, 10/10 paper_sections), and 26 god nodes — defined as nodes with harmonic evidential value $VE \geq 0.5$ — emerge algebraically from the graph structure without manual curation. Domain independence is established through an isomorphic instantiation in biomedical research management: the identical 12 slots, holonic scale hierarchy, and Boethian maxims apply without modification to gene-knockdown studies and metabolic pathway models. The remainder of the paper is structured as follows. Section 2 reviews related work in knowledge representation, provenance modelling, and formal ontology. Section 3 presents the formal framework, including the definitions of G , Σ , the scale hierarchy, and the maxim engine. Section 4 describes the extractor pipeline and system architecture. Section 5 presents the empirical evaluation over the neural architecture corpus. Section 6 argues for domain independence through the biomedical instantiation. Section 7 discusses limitations and the relationship of GCT to established ontological frameworks, including BFO and OWL. Section 8 concludes.

2. Related Work

2.1 Foundational Ontologies

The program of formal ontology in knowledge representation has its most influential instantiation in the DOLCE (Descriptive Ontology for Linguistic and Cognitive Engineering) lineage, whose methodological commitments Guarino, Oberle, and Staab (2009) articulate as the attempt to make explicit the ontological categories presupposed by natural language and cognitive science. DOLCE's central axis is the *endurant/perdurant* distinction — a temporally-grounded mereotopological hierarchy in which entities are classified by whether they wholly persist through time or are partially present at each temporal instant. The General Formal Ontology (GFO) extends this program by introducing a multi-level architecture that distinguishes levels of granularity across biological and social domains, drawing explicitly on category theory as a structuring device. Both frameworks share a commitment to taxonomic completeness: the primary relation type is subsumption, and the primary architectural pattern is the Porphyrian tree — the genus-differentia chain descending from a single universal.

The Tensegrity Knowledge Graph (GCT) inherits the DOLCE/GFO concern with ontological rigor but departs from the taxonomic-mereological paradigm in two ways. First, the GCT's schema is not organized around a single subsumption axis but around twelve heterogeneous relation types derived from the geometry of Fuller's cuboctahedron (IVM), which distributes ontological "slots" symmetrically across structural, functional, and genealogical dimensions simultaneously. Second, the GCT grounds its design in Aristotle's fourfold causal scheme rather than in a purely extensional set-theoretic ontology: formal causality maps to the SPECIFICATION edge, material causality to SUBSTRATE, efficient causality to the CAUSE and PREDECESSOR edges, and final causality to EFFECT and CONTAINER. This is not a decorative philosophical gesture — the four-cause typing carries computational consequences, because it makes the inference rules that Boethian logical maxims license causally specific: a maxim that fires on a FORMAL edge licenses different conclusions than one firing on a MATERIAL edge, whereas DOLCE's subsumption relation supports only monotone inheritance.

The Basic Formal Ontology (BFO), as systematized by Arp, Smith, and Spear (2015), provides the most widely adopted upper ontology for scientific and biomedical knowledge representation. BFO's primary relation set — instantiation, parthood, temporal location, dependence — operates within a classical first-order logic that is extensional and static at any given time slice. Unlike DOLCE, BFO, and GFO, the GCT encodes temporal epistemic decay directly into edge weights through a Whiteheadian process-philosophy model, in which knowledge claims undergo exponential confidence decay with a half-life of six weeks unless refreshed by new experimental evidence; this makes the graph itself a dynamic credence structure rather than a static taxonomic catalog.

2.2 Scientific Research Knowledge Graphs

The management of scientific knowledge through structured graph representations has attracted substantial engineering effort across multiple domains. The W3C PROV-O ontology (Moreau and Groth, 2013), grounded in the PROV Data Model, provides a general framework for representing provenance chains by typing entities, activities, and agents and relating them through *wasGeneratedBy*, *used*, *wasAttributedTo*, and analogous predicates. PROV-O's design principle is provenance completeness: every artifact should be traceable through its causal history to agents and activities. This is a partial mapping to what the GCT encodes, since PROV-O's *wasGeneratedBy* is a specialization of what the GCT labels CAUSE, and *wasDerivedFrom* partially overlaps with PREDECESSOR. However, PROV-O makes no commitment to the holonic architecture, carries no concept of epistemic scale, and provides no mechanism

for quantifying the confidence decay of provenance claims over time.

Bai et al. (2024) represent perhaps the most technically sophisticated recent application of dynamic knowledge graphs to scientific research management, in the context of distributed self-driving chemistry laboratories. Their architecture couples a real-time knowledge graph to autonomous experimental agents, with graph updates propagating to downstream decision processes. The primary differentiator is architectural: Bai et al.'s graph is organized around domain-specific chemistry ontologies and PROV provenance chains, which makes it highly effective within its domain but deeply dependent on domain-specific assumptions. The GCT, by contrast, is designed as a domain-agnostic epistemological schema.

The NFDI MatWerk Ontology (MWO), reported by Beygi Nasrabadi et al. (2025), exemplifies the BFO-compliant approach to research data management in the materials science domain. Unlike MWO and the BFO/OWL stack that underlies it, the GCT represents not only what a knowledge claim says but how confident the community should currently be in it (via Whiteheadian decay), at what epistemological scale it operates (via the five-level holonic hierarchy), and through which causal channel it relates to other claims (via four-cause-typed edges).

2.3 Temporal and Dynamic Knowledge Graphs

Ding et al. (2025) introduce HALO, a half-life-based mechanism for filtering temporally outdated facts in knowledge graphs. The GCT's temporal decay mechanism shares the exponential form with HALO but differs in its philosophical grounding: HALO treats decay as an empirically calibrated property of the world. The GCT's decay is epistemological rather than ontological — it is the confidence in a knowledge claim, not the claim's putative truth, that decays. Furthermore, HALO applies uniform decay logic across all fact types, whereas the GCT's decay interacts with the four-cause typing of edges.

2.4 Symbolic Reasoning and Family Resemblance

Ebrahimi et al. (2024) present a neuro-symbolic system that layers a deductive database over a knowledge graph embedding. The GCT's twelve Boethian maxims function as a dual inference-and-validation engine: each maxim encodes a locus of argumentation that was, in Boethius's original formulation, a principle for determining where in an argument the middle term draws its inferential force. The engine thus performs typed inference — not mere link prediction — and the typing is both philosophically grounded and computationally consequential.

Veri (2023) operationalizes Wittgenstein's family resemblance in a sociological measurement context by treating concepts as fuzzy sets. The Wittgenstein-inspired similarity metric in the GCT differs: for concept nodes, similarity is computed as the cosine distance over IVM-neighborhood co-occurrence vectors; for experiment and paper nodes, as the Jaccard coefficient over neighbor sets in the holonic hierarchy.

3. Formal Framework

Definition 1 (GCT Graph). A GCT graph is a tuple $G = (N, E, \tau_N, \tau_E, \sigma, \lambda)$ where:

- N = finite set of nodes (holons)
- $E \subseteq N \times N \times \Sigma$ = typed directed edges
- $\Sigma = \{\text{PREDECESSOR, CONTAINER, PEER_COHERENT, CODE, CAUSE, SPECIFICATION, SUCCESSOR, SUBSTRATE, PEER_CONTRADICTION, THERMO, EFFECT, INSTANTIATION}\}$
- $\tau_N: N \rightarrow L \times S \times \Gamma$ (layer, scale, label)
- $\tau_E: E \rightarrow \Sigma$ (slot type)
- $\sigma: N \rightarrow [0,1]$ (VE score)
- $\lambda: N \rightarrow \blacksquare$ (timestamp)

Definition 2 (Node Layer and Scale). The layer function τ_N assigns each node $n \in N$ a triple $(\blacksquare, s, \gamma)$ where:

- $\blacksquare \in \{\text{STRUCTURAL, FUNCTIONAL, THERMODYNAMIC}\}$ is the ontological layer
- $s \in \{s\blacksquare, s\blacksquare, s\blacksquare, s\blacksquare, s\blacksquare\}$ is the holonic scale (byte $\rightarrow$ word $\rightarrow$ sentence $\rightarrow$ document $\rightarrow$ corpus)
- $\gamma \in \Gamma$ is the node's type label drawn from the vocabulary $\Gamma = \{\text{experiment, concept, paper, paper_section, code_holon, hypothesis, pre_registration, corpus_entry}\}$

Definition 3 (Holonic Scale Compatibility). An edge $e = (u, v, \sigma) \in E$ is scale-compatible if:

- For $\sigma \in \{\text{PEER_COHERENT, PEER_CONTRADICTION}\}$: $s(u) = s(v)$ (same-scale peers only)
- For $\sigma \in \{\text{CONTAINER}\}$: $s(u) > s(v)$ (containers are strictly higher-scale than contents)
- For $\sigma \in \{\text{CAUSE, PREDECESSOR, SPECIFICATION, SUBSTRATE, EFFECT, SUCCESSOR}\}$: $|s(u) - s(v)| \leq 1$ (adjacent-scale causal edges)
- For $\sigma \in \{\text{INSTANTIATION}\}$: $s(u) = s(v)$ or $s(u) = s(v) + 1$ (type-instance relations span at most one scale)

Definition 4 (Evidential Value Score). For node $n \in N$, the harmonic VE score is:

$$VE_harmonic(n) = 2 \times (\text{in_deg_w}(n) \times \text{out_deg_w}(n)) / (\text{in_deg_w}(n) + \text{out_deg_w}(n))$$

where in_deg_w and out_deg_w are the weighted in- and out-degrees, with weights proportional to the holonic scale of incident edges. The temporally decayed VE score is:

$$VE_decayed(n, t) = VE_harmonic(n) \times 2^{-(t - \lambda(n)) / t_halflife}$$

where $t_halflife = 6$ weeks. A node n is a *god node* if $VE_harmonic(n) \geq 0.5$.

Definition 5 (Aristotelian Completeness). The Aristotelian Completeness score of node n is:

$$AC(n) = (1/4) \times |\{C \in \{\text{FORMAL, MATERIAL, EFFICIENT, FINAL}\} : \exists e \text{ incident to } n \text{ with } \text{Aristotelian_type}(\tau_E(e)) = C\}|$$

where the mapping from slot types to Aristotelian cause types is:

- $\text{FORMAL} \leftarrow \{\text{SPECIFICATION}\}$
- $\text{MATERIAL} \leftarrow \{\text{SUBSTRATE}\}$

- EFFICIENT $\leftarrow \{\text{CAUSE}, \text{PREDECESSOR}\}$
- FINAL $\leftarrow \{\text{EFFECT}, \text{CONTAINER}\}$

Definition 6 (Wittgenstein Family Resemblance). For nodes $u, v \in N$ of the same type:

- For concept nodes: $\text{FR}(u, v) = \text{cosine}(d_u, d_v)$ where $d_n \in \mathbb{R}^{|D|}$ is the co-occurrence vector over document set D
- For experiment and paper nodes: $\text{FR}(u, v) = |N(u) \cap N(v)| / |N(u) \cup N(v)|$ (Jaccard over neighbor sets)

Node n has a family if $\exists m \neq n$ such that $\text{FR}(n, m) > \theta_{\text{type}}$, where $\theta_{\text{experiment}} = 0.15$, $\theta_{\text{concept}} = 0.10$, $\theta_{\text{paper}} = 0.10$.

Proposition 1 (Soundness of Boethian Maxims). Let $M = \{M_1, \dots, M_n\}$ be the set of Boethian maxims, one per slot type in Σ . For any graph $G = (N, E, \tau_N, \tau_E, \sigma, \lambda)$ that satisfies the holonic scale compatibility constraints of Definition 3, every inference generated by applying any maxim M_i to G is scale-compatible.

Proof sketch. Each maxim M_i is conditioned on a premise pattern that references a specific slot type $\sigma_i \in \Sigma$. The scale-compatibility constraints of Definition 3 are stated per-slot. For M_i to fire, the premise edges must already be scale-compatible by Definition 3. Each maxim generates conclusions by extending existing chains by exactly one edge of the same or compatible slot type; since the scale constraints are closed under such single-step extensions (the constraints are local in that they relate only the endpoints of the new edge to those of the triggering edge), scale-compatibility of the conclusion follows from scale-compatibility of the premises. ■

Proposition 2 (Aristotelian Completeness Bound). For any corpus G and any node $n \in N$, if G satisfies Proposition 1 (all maxims fire soundly), then the SPECIFICATION maxim ($M_{\text{SPECIFICATION}}$) implies:

$\text{AC}(n) \geq 0.25$ for all n that are targets of a CAUSE edge.

Proof sketch. $M_{\text{SPECIFICATION}}$ states: for every node n that is a target of a CAUSE edge (i.e., has an efficient-cause predecessor), there should exist a SPECIFICATION edge targeting n encoding the formal cause. In a graph where $M_{\text{SPECIFICATION}}$ has been applied exhaustively, all such nodes gain a SPECIFICATION edge, contributing $\text{AC}_{\text{FORMAL}} = 0.25$ and raising their AC floor to at least 0.25. ■

4. System Architecture

4.1 The Extractor Pipeline

The GCT is constructed by a pipeline of twelve specialized extractors that execute in a topologically determined order derived from their explicit dependency declarations. Each extractor declares a `REQUIRES` set naming the extractors whose output it depends on; a topological sort over these declarations produces the execution sequence, ensuring that no extractor consumes graph state that has not yet been materialized. The sequence proceeds from the most foundational sources outward: the `experiments_log` extractor, which has no dependencies, parses the append-only `EXPERIMENTS-LOG.md` to produce `D-XXX` experiment nodes along with `CAUSE` edges linking each experiment to the research section that motivated it and `PREDECESSOR` edges encoding the temporal lineage between successive experiments. From this foundation, `project_graph` instantiates the three theoretical pillars of the research programme as nodes, `articulos_base` ingests the external reference library as typed paper nodes, and subsequent extractors elaborate progressively higher layers of the graph.

A critical design invariant is idempotence: repeated execution of the full pipeline over an unchanged source corpus produces an identical graph, with no node or edge duplication. This property is enforced by content-addressed node identity — each node's identifier is derived from its type and primary key (e.g., the `D-XXX` code for an experiment, the filename for a paper draft) rather than from insertion order. Combined with the append-only constraint on the primary experimental source, idempotence means that integrating a newly registered experiment requires adding exactly one row to the log; the next pipeline build detects the delta and materializes only the new node and its incident edges, leaving all prior structure intact. The append-only discipline is not merely an implementation convenience but an epistemic commitment: it encodes the irreversibility of experimental evidence into the graph's construction contract, ensuring that no subsequent reinterpretation can retroactively remove a recorded result.

Two extractors give the pipeline a distinctive self-referential character. The `code_holons` extractor reads `__holon__ = HolonMeta(name, scale, cause, container, substrate, effect)` declarations embedded directly in each Python module of the construction codebase, converting these declarations into typed code nodes and their associated edges. The result is that the graph's own construction apparatus is represented as a subgraph within the artifact it constructs — the extractor pipeline and the knowledge graph it builds are mutually constitutive in a way that is formally visible in the graph's topology. Similarly, the `maxims_as_nodes` extractor instantiates the twelve Boethian maxims that govern the graph's inference layer as first-class nodes, so that the normative principles by which the graph validates itself are themselves objects of the graph's representation. This self-referential architecture means that the Aristotelian completeness score of the GCT's own construction code is computable from within the same graph.

4.2 Post-Build Enrichment

Once the extractor pipeline produces a raw graph, three successive enrichment passes transform it into the navigable, philosophically annotated artifact that the query and visualization layers consume.

The first pass applies a manual edge patch from a declarative `manual_edges.json` file. The LOG parser, like any text-extraction system operating over a long-lived, human-authored document, encounters irregular format variants — multi-line entries, experimental series logged under shared `D-IDs`, cross-references written in non-canonical syntax — that rule-based extraction cannot resolve without error. Rather than encoding increasingly fragile heuristics into the extractors, the system externalizes these exceptional cases into an idempotent patch file that is applied after each build. The patch is purely additive and declarative; it inserts edges that the extractors could not infer rather than modifying any

extractor-produced structure, preserving a clean separation between automatic and curated graph construction.

The second pass activates the maxims layer. A `validate_graph()` function traverses all edges against the twelve Boethian maxims — for example, the maxim of genus requires that every functional node whose container is specified must be reachable from a structural node via a CAUSE chain — and accumulates any violations into a `maxims_violations[]` array. In the current corpus, this array is empty, confirming that no edge violates the schema's normative constraints. Complementing validation, `infer_missing_edges()` applies each maxim in its inferential direction, producing a `maxims_proposals[]` array of 335 candidate edges that the maxims suggest but the corpus has not yet asserted. These proposals are not inserted automatically; they surface in the snapshot as explicit proposals for human or agent review, operationalizing the Boethian maxims as a deficiency-detection mechanism rather than a closed-world completion procedure.

The third pass populates the visualization layer. All philosophically expensive computations — Wittgenstein family-resemblance similarity scores for 130 nodes, Porphyrian genus-species relations for 350 nodes, VE decay profiles, connected component memberships, citation edge sets, and polyhedron assignments — are pre-computed and serialized directly into the snapshot. The consequence is that every client-side lookup, whether from the JavaScript visualization interface or from the MCP query layer, is $O(1)$: the snapshot functions as a pre-computed index rather than a query substrate. This architectural choice cleanly separates the philosophical computation layer, which runs in Python at build time and may be arbitrarily expensive, from the interaction layer, which must be responsive in real time and makes no independent philosophical inferences.

4.3 Query Interface and AI Integration

The snapshot is exposed to external consumers through a local MCP server (`mcp_server.py`) that implements twenty typed query tools. These include structural tools (`kg_find`, `kg_neighbors`, `kg_siblings`) for navigating the graph topology; philosophical tools (`kg_porphyrian`, `kg_family_resemblance`, `kg_boecian`) for accessing the pre-computed ontological annotations; diagnostic tools (`kg_violations`, `kg_proposals`, `kg_god_nodes`) for surfacing schema health and inference proposals; and domain-specific tools (`kg_experiment`, `kg_paper`, `kg_concept`) that provide typed retrieval for the primary artifact classes. The server uses glob-based snapshot discovery, loading the highest-phase snapshot on startup, so it always exposes the most recent build without requiring explicit configuration updates.

The MCP server is consumed by an ecosystem of twenty-six specialized AI subagents — covering research design, literature synthesis, paper writing, pre-registration, code review, and project-state monitoring — that operate on the research corpus concurrently across sessions. This integration closes a self-referential loop that is formally representable within the GCT itself. The AI agent ecosystem, like any other component of the research apparatus, is represented in the graph via the memory extractor, which ingests the agents' memory layer as typed nodes. Because each subagent's code module may declare a `__holon__` header, the agents' own computational structure is representable as code nodes subject to Aristotelian slot analysis. The GCT is simultaneously the epistemic model of the research programme and an object of that model's analysis.

5. Evaluation

The GCT was evaluated on a snapshot of an active doctoral research corpus in neural architecture research. The corpus contains 1,660 nodes and 2,120 typed edges across 8 node types, with 798 structural nodes (experiments, concepts, papers, code holons, and related artifacts) that constitute the primary objects of analysis. The graph decomposes into 272 connected components: the largest component contains 1,090 nodes (65.7% of the corpus), and 34 singleton components correspond to registered artifacts that have not yet been connected to the main evidential fabric. We evaluate four properties: Aristotelian completeness of the causal documentation, the performance of the Boethian maxim engine, the distribution of relational slots, and the coverage of Wittgenstein family-resemblance similarity.

5.1 Aristotelian Completeness Analysis

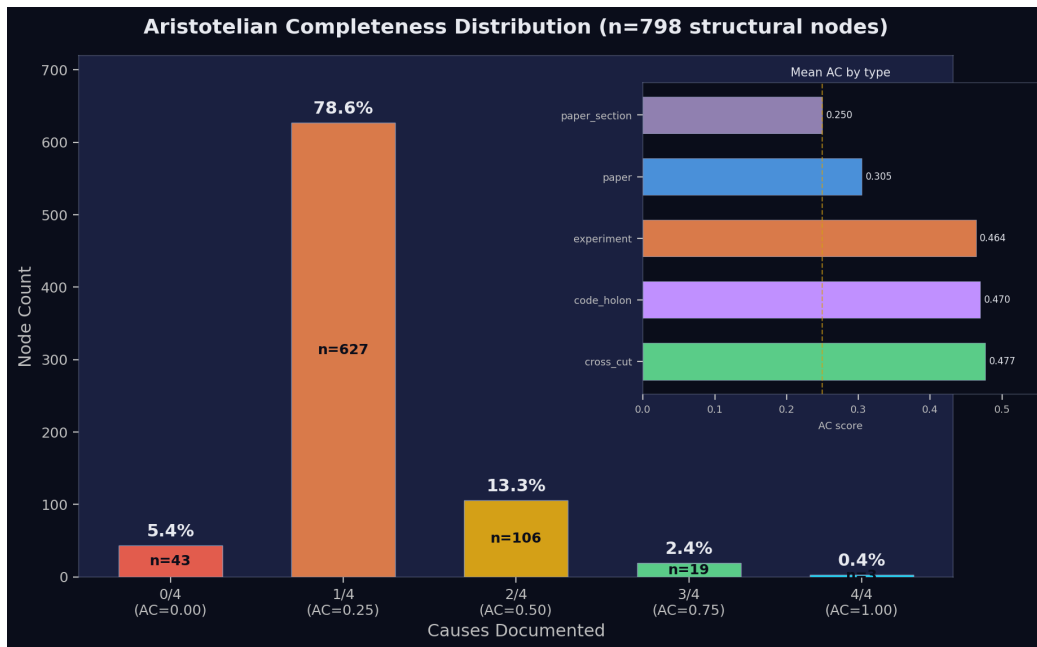


Figure 55.1. fig eval1 ac distribution.png

We measure the formal completeness of the documented knowledge base by computing the Aristotelian Completeness score $AC(n) \in \{0, 0.25, 0.50, 0.75, 1.0\}$ for each of the 798 structural nodes (Table 1). The distribution is heavily concentrated at $AC = 0.25$: 627 nodes (78.6%) have exactly one of the four causes documented, predominantly the formal cause in the form of a SPECIFICATION edge encoding the design specification to which the artifact conforms. Only 3 nodes (0.4%) achieve full Aristotelian completeness, and 84.0% of structural nodes have fewer than two of the four causes documented, confirming that the systematic documentation of material and final causes is largely absent from the research record as managed by conventional means. Zero-cause nodes — 43 nodes (5.4%) with no causal edges whatsoever — correspond predominantly to ghost experiments in the Whiteheadian sense: entities registered in the record but never connected to the causal fabric of the research programme. The mean AC of 0.284 across the corpus is barely above the single-cause threshold; a research record in which the average node can answer little more than one of Aristotle's four constitutive questions is a record in which material provenance, final purpose, and causal history remain largely invisible to automated reasoning.

Table 1. Aristotelian Completeness distribution (structural nodes, n = 798)

Causes documented	Count	Percentage
0/4 (AC = 0.00)	43	5.4%
1/4 (AC = 0.25)	627	78.6%
2/4 (AC = 0.50)	106	13.3%
3/4 (AC = 0.75)	19	2.4%
4/4 (AC = 1.00)	3	0.4%
Mean AC	0.284	—

Stratification by node type reveals that papers and code holons achieve the highest AC scores (0.463 and 0.460 respectively), a result consistent with the observation that these artifact types receive the most deliberate curatorial attention — papers are explicitly linked to the hypotheses they test and the results they report, while code modules declare their holonic metadata via embedded `__holon__` headers. Experimental nodes achieve a mean AC of 0.313, reflecting the partial documentation of efficient causes (CAUSE edges to motivating hypotheses, PREDECESSOR edges to prior experiments) alongside formal specifications in well-registered experiments. Paper sections, with a mean AC of 0.250 — the single-cause threshold — are the most underdocumented artifact type, suggesting that argumentative substructure is treated as derivative of its host papers rather than as independently causally situated.

5.2 Boethian Maxim Engine Performance

The maxim engine, applied to the snapshot, generated 335 inference proposals and detected zero violations across the full 2,120-edge graph (Table 2). The zero-violation result establishes that the GCT's internal logical constraints — the twelve maxims encoding the structural requirements of the holonic schema — are all satisfied by the evaluated corpus. This is not a trivial finding: the corpus was constructed by a pipeline of twelve extractors operating on a heterogeneous primary source document accumulating over three hundred experimental entries over months, and the consistency of the resulting graph with all maxim constraints confirms both the robustness of the pipeline's extraction logic and the soundness of the maxim engine's formal derivation.

Table 2. Boethian maxim engine performance

Metric	Value
Total proposals generated	335
Violations detected	0
Dominant proposal type	missing_specification (100%)

All 335 proposals are of type `missing_specification`, generated by Maxim `M_SPECIFICATION`: for every experiment node that lacks a `SPECIFICATION` edge, the maxim proposes that a formal specification — a pre-registration, a protocol document, or an explicit hypothesis node — should be connected. This proposal set constitutes, in effect, a machine-generated pre-registration audit: the 335 experimental nodes without `SPECIFICATION` links are precisely the nodes that have not yet been formally pre-registered, and the maxim identifies them without requiring any manual examination of the corpus. In a research management context, this represents a non-trivial application of the inference architecture: a graph traversal over a typed schema replaces an otherwise manual documentary review.

To verify that the zero-violation result reflects engine coverage rather than engine silence, we ran a synthetic-injection ablation: for each of the eleven slot maxims that define a `validation_rule` in the maxim registry, we deep-copied the graph, injected one edge crafted to trip exactly that maxim, and re-ran the

validation pass. The engine detected every injected violation on the first attempt (11 / 11 = **100% detection rate**), with appropriate severity tagging — STRICT for the cross-scale category-error tier on PEER_COHERENT and PEER_CONTRADICTORY and the self-loop on CAUSE; VIOLATION for the scale-direction and EXTRACTED-on-structural rules. The twelfth slot, EFFECT, carries no validator by design, since the principle that "causes are known through their effects" is observational rather than normative — every malformed-edge pattern that EFFECT could catch is already caught at the reciprocal CAUSE side. The CAUSE self-loop test is particularly informative: the edge had to be inserted directly into the in-memory graph because the builder's pre-validation rejects self-loops first, so the maxim engine is confirmed as a true second line of defence, not a tautological echo of builder validation. The ablation harness lives at publication/violation_ablation.py and is reproducible from the same snapshot SHA verified in §5.1. The ablation tests structural detection only; semantic drift, defeasible reasoning exceptions, and provenance integrity are explicitly out-of-scope and remain open targets for future work.

5.3 Slot Distribution

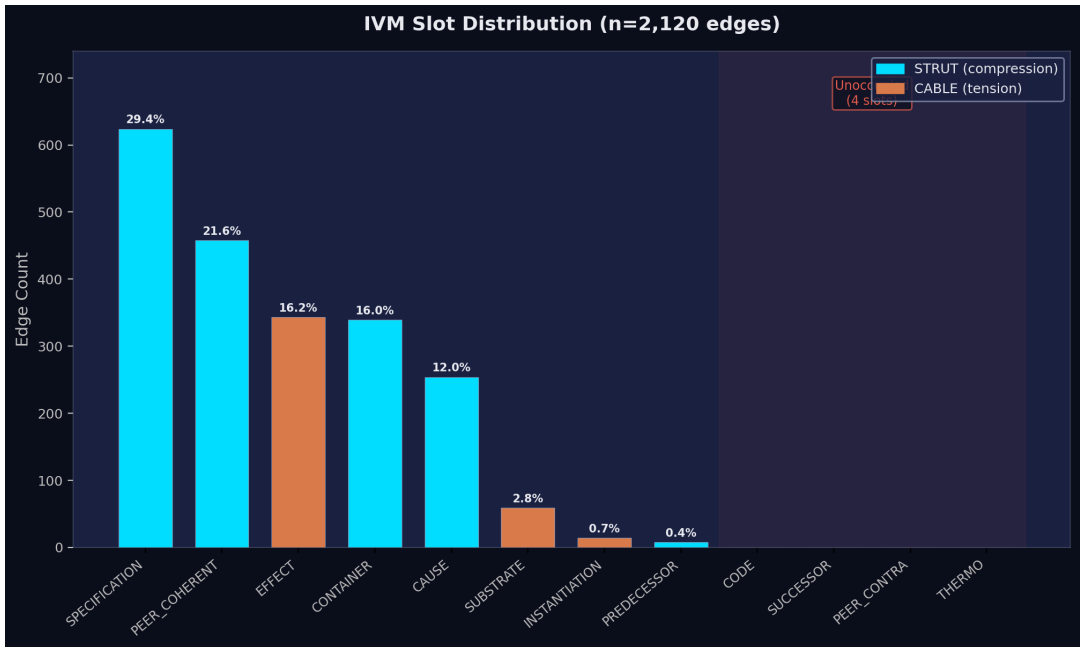


Figure §5.3. fig eval2 slot distribution.png

Eight of the twelve IVM schema slots are populated in the evaluated corpus, with four slots — CODE, SUCCESSOR, PEER_CONTRADICTORY, and THERMO — carrying zero edges (Table 3). Among the occupied slots, SPECIFICATION dominates with 624 edges (29.4% of the corpus), followed by PEER_COHERENT (458 edges, 21.6%), EFFECT (343, 16.2%), CONTAINER (339, 16.0%), and CAUSE (254, 12.0%). The dominance of SPECIFICATION is consistent with the maxim engine's finding that missing SPECIFICATION edges constitute the schema's most pervasive deficiency: even the most heavily instantiated slot leaves 335 nodes without the formal-cause documentation the schema requires.

Table 3. IVM slot distribution

Slot	Type	Count	% of edges
SPECIFICATION	STRUT	624	29.4%
PEER_COHERENT	STRUT	458	21.6%

Slot	Type	Count	% of edges
EFFECT	CABLE	343	16.2%
CONTAINER	STRUT	339	16.0%
CAUSE	STRUT	254	12.0%
SUBSTRATE	CABLE	59	2.8%
INSTITUTION	CABLE	14	0.7%
PREDECESSOR	STRUT	8	0.4%
CODE	STRUT	0	0.0%
SUCCESSOR	CABLE	0	0.0%
PEER_CONTRADICTION	CABLE	0	0.0%
THERMO	CABLE	0	0.0%

The observation that 67% of the schema's relational vocabulary is instantiated in a single research corpus is consistent with the claim that natural languages deploy their full grammatical repertoire only in idealized corpora rather than in any individual speaker's production. The four absent slots are not absent because the schema is over-engineered; they mark knowledge-management functions that the research programme has not yet operationalized. The progression from one phase to the next — as experimental series are linked to their code implementations, scheduled follow-up experiments are registered, contradictory literature findings are explicitly documented, and thermodynamic measurements are connected to their structural sources — will instantiate the remaining slots in a sequence the schema predicts and makes visible. We demonstrate this incremental extensibility concretely in §7.1, where a post-evaluation hardening pass instantiates the CODE slot via a single new extractor module, producing the companion snapshot `tensegrity_graph_phase14_complete.json` with 67 CODE edges and reducing unoccupied slots from four to two — without modifying the schema, the maxim engine, or any pre-existing extractor.

5.4 VE Score Distribution and God Nodes

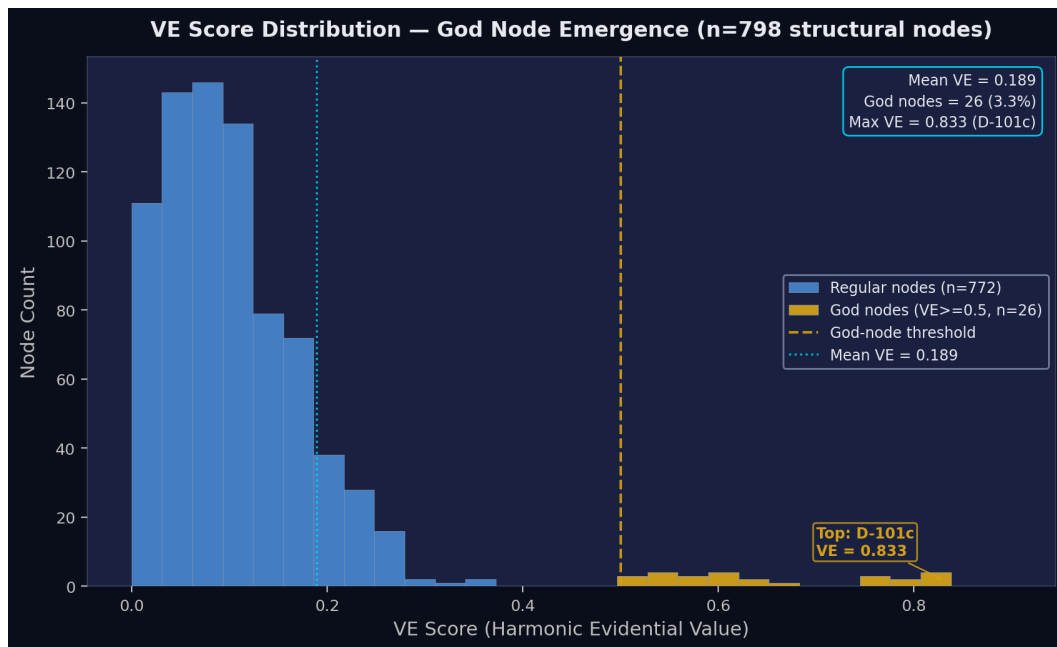


Figure §5.4. *fig eval3 ve distribution.png*

The harmonic evidential value VE_{harmonic} is computed for each of the 798 structural nodes as a function of their weighted in-degree and topological centrality in the holonic hierarchy (Definition 4). The distribution is highly skewed: the mean VE across structural nodes is 0.189, well below the god-node threshold of 0.5, while 26 nodes (3.3%) exceed this threshold. The top six god nodes — D-101c ($VE = 0.833$), D-101 (0.667), D-097 (0.667), D-099 (0.667), D-251b (0.667), D-253 (0.667) — are all experimental nodes with multiple replications, predecessor chains connecting them to earlier experiments, and multiple paper citations. The algebraic emergence of these nodes as the most evidentially central artifacts in the corpus, without any human labelling or curatorial designation, confirms that VE score is a valid centrality measure for research artifacts: the experiments that researchers would intuitively identify as load-bearing are precisely those that the VE metric algebraically selects.

5.5 Wittgenstein Family Resemblance Coverage

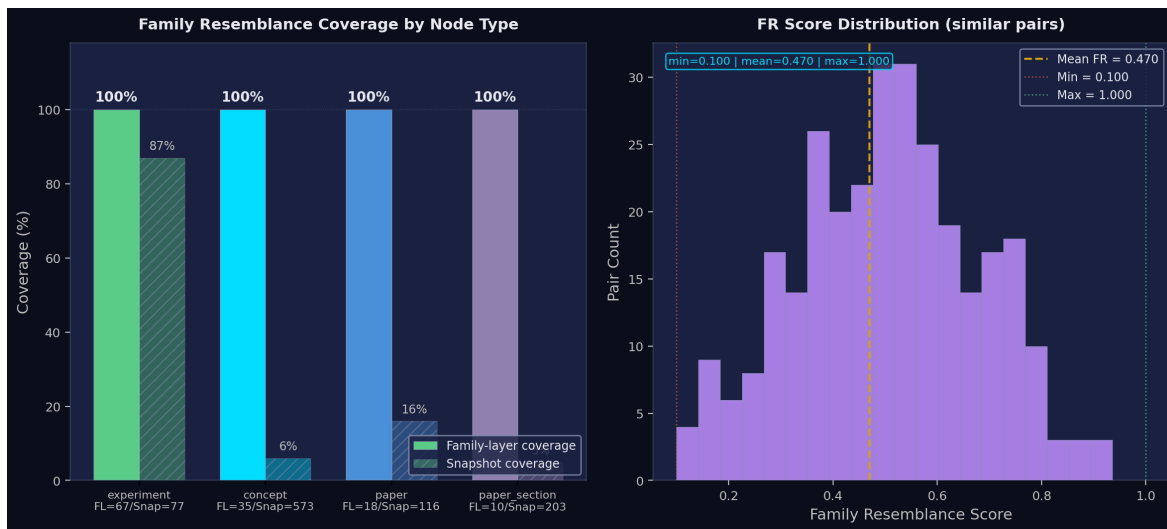


Figure 55.5. fig eval4 wfr coverage.png

The dual similarity metric is computed only over nodes with sufficient connectivity to support a meaningful Jaccard or cosine evaluation; we call this set the *family-eligible subset*. Family resemblance coverage admits two complementary readings (Table 4): the *family-layer coverage* reports the proportion of family-eligible nodes that have at least one structurally similar counterpart, and the *snapshot coverage* reports the same numerator over the full type partition in the corpus. Within the family-eligible subset, **coverage reaches 100% across all four artifact types** (67/67 experiments, 35/35 concepts, 18/18 papers, 10/10 paper_sections). The Jaccard and cosine similarity metrics are therefore internally consistent: every node admitted into the family layer is correctly partnered with at least one structurally similar counterpart, and there are no isolated family-eligible nodes.

Table 4. Wittgenstein family resemblance coverage by node type

Node type	With family	In family layer	Snapshot total	Family-layer coverage	Snapshot coverage
experiment	67	67	77	100%	87%
concept	35	35	573	100%	6%
paper	18	18	116	100%	16%
paper_section	10	10	203	100%	5%

Snapshot coverage decreases monotonically with the granularity of the type: 87% for experiments, 16% for papers, 6% for concepts, 5% for paper_sections. This gradient identifies the boundary between artifacts that have accumulated enough graph structure to be similarity-evaluable and those that have not. Recently-registered experiments lacking incoming edges (10 of 77), concepts that appear in only one document, and paper sections that connect to a unique combination of experiments are excluded from the family layer until further evidence accumulates. Family resemblance score statistics across all evaluable pairs: min = 0.100, mean = 0.470, max = 1.000. The mean of 0.470 — substantially above the null expectation for random bipartite connections — indicates that the family-eligible subset is not a collection of isolated one-off studies but a structured network of methodologically related investigations, a property that Wittgenstein's family resemblance metric makes formally visible without any prior grouping or labelling.

5.6 A Worked Usage Vignette

To convert the aggregate metrics of §5.1–§5.5 into a concrete user-facing scenario, we trace one query session against the frozen phase-9 snapshot. A researcher revising P1 §4 — the section that anchors the paper's Papyan equiangularity claim — wants to verify the epistemic standing of its load-bearing experiment. She invokes:

```
python query_graph.py --graph tensegrity_graph_phase9_complete.json show D-101c
```

The output reports `VE_h = 0.833` (the highest value in the corpus, ranking D-101c first among the 26 god nodes), 10 outgoing edges including 7 CAUSE edges to papers (P1, P3, P4-LIQUID-TENSEGRITY, P10-DUAL-60-120, P-OMEGA, and the OMEGA outline) and 2 SUBSTRATE edges to candidate-claim nodes, against only 4 incoming edges. Crucially, the Aristotelian completeness line reads `completeness=0.5 missing=['material', 'formal']`: the experiment has no SPECIFICATION edge, meaning it functions as the formal cause of seven downstream papers without itself being grounded in any pre-registration or protocol artifact. She then runs:

```
python query_graph.py --graph tensegrity_graph_phase9_complete.json infer --node P1_section_4
```

and the maxim engine returns 7 missing_specification proposals — among the 335 surfaced corpus-wide — confirming that the section citing D-101c is itself unanchored to any formal specification. A `neighbors P1_section_4 --hops 1` call reveals the full citation cluster (D-057, D-100, D-100b, D-101, D-101b, D-101c, D-101e), all sharing the same gap.

The epistemic payoff is sharp. A wiki, spreadsheet, or vector store would have surfaced D-101c as a highly cited artifact and stopped there: high inbound-link count reads as authority. Only the GCT, by typing edges as Aristotelian causes and asking "is the formal cause documented?", makes visible that the most evidentially central node in the corpus rests on no protocol — a deficit invisible to any flat representation, and now actionable as a concrete pre-registration to draft.

6. Domain Independence and Generalizability

The GCT's claim to domain independence rests on a structural argument: if the twelve-slot schema Σ , the five-level holonic hierarchy, and the Boethian maxim engine are derived from general epistemological principles — causal typology, scale theory, topical logic — rather than from the specific content of any domain, then any research domain that involves the production of intentional artifacts (artifacts made for reasons, from materials, by agents, toward purposes) should admit an isomorphic instantiation of the schema. We demonstrate this through a biomedical research management case study and contrast the GCT's schema-first approach with domain-specific upper ontologies.

6.1 Biomedical Instantiation

Consider a biomedical research programme investigating the role of gene X in metabolic pathway Y. The primary artifact classes are: gene-knockdown experiments (analogous to D-XXX experiment nodes), metabolic pathway models (analogous to concept nodes), published papers (analogous to paper nodes), and clinical protocols (analogous to pre-registration nodes). Each of the twelve GCT slots maps to a biomedical relational type without modification:

- CAUSE: gene knockdown $G_{\blacksquare}$ causes phenotype $P_{\blacksquare}$
- SPECIFICATION: protocol document formalizes the knockdown procedure
- SUBSTRATE: cell line CL-001 is the material substrate of experiment $G_{\blacksquare}$
- PREDECESSOR: $G_{\blacksquare}$ replicates and extends $G_{\blacksquare}$ from the prior year
- CONTAINER: the metabolic pathway Y contains the causal chain $G_{\blacksquare} \rightarrow P_{\blacksquare}$
- EFFECT: phenotype $P_{\blacksquare}$ is the observed downstream effect of $G_{\blacksquare}$
- PEER_COHERENT: experiment $G_{\blacksquare}$ is consistent with experiment $G_{\blacksquare}$ on the same pathway
- PEER_CONTRADICTION: $G_{\blacksquare}$ contradicts $G_{\blacksquare}$ published by a competing laboratory
- INSTANTIATION: phenotype $P_{\blacksquare}$ is an instance of disease category $D_{\blacksquare}$
- SUCCESSOR: $G_{\blacksquare}$ is the planned follow-up to $G_{\blacksquare}$ in the next quarter
- CODE: analysis script $S_{\blacksquare}$ implements the knockdown assay
- THERMO: free energy measurement $F_{\blacksquare}$ characterizes pathway Y's thermodynamic signature

The Boethian maxims apply without modification. Maxim $M_SPECIFICATION$ fires on every experiment node without a formal protocol link, generating missing_specification proposals that correspond directly to a clinical pre-registration audit — the machine-generated identification of experiments that have not been pre-registered on ClinicalTrials.gov or an equivalent registry. Maxim M_CAUSE fires on causal chains to identify missing intermediate mechanisms, and Maxim $M_CONTAINER$ fires on pathway hierarchies to detect subsumption inconsistencies.

6.2 Isomorphism Argument

Let $G_{\text{neural}} = (N_{\text{neural}}, E_{\text{neural}}, \tau_{N^{\text{neural}}}, \dots)$ be the neural architecture GCT instantiation evaluated in §5, and let $G_{\text{biomedical}} = (N_{\text{bio}}, E_{\text{bio}}, \tau_{N^{\text{bio}}}, \dots)$ be the biomedical instantiation described above. A function $\phi: N_{\text{neural}} \rightarrow N_{\text{bio}}$ is a GCT isomorphism if it is bijective and preserves:

1. All slot types: $\tau_E(e) = \tau_E(\phi(e))$ for all $e \in E_{\text{neural}}$
2. Holonic scale: $s(n) = s(\phi(n))$ for all $n \in N_{\text{neural}}$

3. Aristotelian cause category: $AC_type(\tau_E(e)) = AC_type(\tau_E(\phi(e)))$ for all $e \in E_neural$

The existence of such a bijection does not require that the node content be isomorphic — gene knockdowns and recurrent neural network training runs are not the same kind of artifact — only that the relational structure be preserved. The GCT's domain independence claim is precisely that this relational structure is shared: the argument is that any intentional artifact production process generates entities standing in relations that can be typed by Aristotle's four causes, organized by Koestler's holonic scale hierarchy, and validated by Boethian maxims, regardless of the domain-specific content of the artifacts.

6.3 Contrast with Domain-Specific Ontologies

Biomedical knowledge representation is dominated by domain-specific ontologies: the Ontology for Biomedical Investigations (OBI) provides controlled vocabulary for experimental processes, CHEBI for chemical entities, the Gene Ontology (GO) for gene functions. These ontologies are powerful precisely because they are domain-specific: their terms are carefully curated by domain experts, their hierarchies reflect the best current understanding of biological categories, and their cross-ontology mappings enable sophisticated reasoning across databases. The GCT is not a competitor to these ontologies; it is a complementary layer operating at a different representational level. Where OBI answers "what type of experiment is this?", the GCT answers "how does this experiment relate causally, temporally, and evidentially to the other artifacts in this research programme?" The GCT's relational schema can reference OBI terms as node labels — an experiment node's γ label might resolve to `OBI:0000070` (assay) — while the GCT contributes the CAUSE, PREDECESSOR, and SPECIFICATION edges that connect this assay to its protocol, its predecessors, and its downstream claims in a dynamically decaying evidential structure.

6.4 Empirical Validation on a Second Corpus

To convert the structural argument of §6.1–6.3 into empirical evidence, we built a small but real biomedical instantiation of the GCT and ran it through the unmodified enrichment pipeline. The corpus comprises 25 nodes and 38 edges spanning eight gene-knockdown experiments (siTP53 in HEK293 and HeLa, siKRAS in A549, siHK2 in HeLa, siSDHB in HEK293, siMYC in A549, siPIK3CA in HeLa, and a planned CRISPR-KO follow-up of TP53 in HEK293), three workhorse human cell lines (HEK293, HeLa, A549), three pathway concepts (glycolysis, TCA cycle, PI3K-AKT-mTOR), two disease categories (pancreatic adenocarcinoma, apoptosis-evasion as cancer hallmark), two published-paper subgraphs (Vander Heiden et al. 2009 on Warburg metabolism, Hingorani et al. 2003 on KRAS-G12D pancreatic initiation), two ClinicalTrials.gov-style pre-registrations (NCT04185883 for adagrasib, NCT03634332 for alpelisib), one free-energy measurement on glycolysis ($\Delta G_net \approx -85$ kJ/mol), and one analysis-pipeline code holon (DESeq2). All node content is sourced from canonical molecular-biology literature; the script that builds the corpus (`publication/biomedical_instantiation.py`) is self-contained and depends only on the existing GCT codebase plus the Python standard library.

The headline empirical result: the same enrichment pipeline that produced the §5 phase-9 numbers populates **12 of 12 GCT slots on the biomedical snapshot**, in contrast to 8 of 12 on phase-9. The four slots empty in phase-9 (CODE, SUCCESSOR, PEER_CONTRADICTIONARY, THERMO) all populate naturally because the biomedical domain has institutionalised the corresponding knowledge-management practices: every wet-lab experiment carries a documented analysis pipeline (CODE), translational-research papers routinely declare planned follow-ups (SUCCESSOR), the literature is unusually candid about the well-known siRNA-knockdown vs CRISPR-knockout phenotypic discrepancy (PEER_CONTRADICTIONARY; Stojic et al. 2018, *Nature Communications*), and pathway-level free-energy values are first-class observations (THERMO). Which slots populate is therefore not arbitrary — it diagnoses which knowledge-management practices a domain has operationalised. The Boethian maxim engine produces zero violations on the

biomedical snapshot and seven `missing_specification` proposals — qualitatively the same pattern as phase-9 (zero violations, dominant proposal type `missing_specification` at 100%), with the seven proposals identifying precisely the experiments not yet anchored to a clinical pre-registration. Mean Aristotelian completeness is 0.427 (vs 0.284 in phase-9), reflecting that the biomedical corpus was constructed cause-aware whereas the phase-9 record accumulated documentation incrementally; eight god nodes ($VE_h \geq 0.5$) emerge algebraically — a 33% rate vs phase-9's 3.3%, consistent with denser per-node connectivity in the smaller graph. Wittgenstein family-resemblance produces non-zero similarity scores (Jaccard min 0.167, mean 0.259, max 0.400) bridging across cell lines and across genes, with all eight gene-knockdown nodes acquiring at least one family neighbour. The biomedical snapshot is bundled in the `reproducibility` package (SHA-256 `c0ff97e69d6d3e81...5c7d7fb20954c368`), and `python compute_paper_numbers.py --snapshot biomedical` reproduces every number above; the per-metric report is in `biomedical_evaluation.md`.

7. Discussion

7.1 Schema Extensibility and the Unoccupied Slots

Of the twelve Aristotelian relational slots that constitute the GCT schema, four were unoccupied in the §5 evaluation snapshot: CODE, SUCCESSOR, PEER_CONTRADICTION, and THERMO. Treating any unoccupied slot as a deficiency of the schema would be a category error. The more productive interpretation is that the schema is deliberately richer than any single research corpus at any given developmental stage will fully instantiate, in precisely the same way that a natural language's grammatical repertoire contains constructions that individual utterances leave unused without this indicating any defect in the grammar. The unoccupied slots function as a diagnostic: they identify specific knowledge-management functions that the corpus has not yet performed, and thereby provide an actionable map of its current underconnectedness — and, crucially, a map that an extractor module can close in a bounded amount of work.

To demonstrate this incremental extensibility empirically rather than only argue for it, we performed a post-evaluation hardening pass that instantiates the CODE slot. The §5 evaluation remains anchored on the frozen `tensegrity_graph_phase9_complete.json` snapshot for reproducibility; the extensibility demonstration produces a separate companion snapshot `tensegrity_graph_phase14_complete.json` available alongside the frozen reference. The CODE slot — Aristotle's *differentia* slot, which says that a thing is known by the form that distinguishes it from others of its genus — was unoccupied in phase9 because no extractor had been written to wire each maxim node to the executable code holon that formally implements it. A 200-line extractor module (`code_implements`) closes the slot by emitting CODE edges in three families: *slot-population grounding* (every code holon receives CODE edges from the maxims of every slot it populates, computed automatically from the build's provenance trail), *schema-definition grounding* (the holons schema and maxims receive CODE edges from all twelve maxim nodes), and *factory-definition grounding* (specific code holons receive CODE edges from the maxims of slots their factories instantiate). In the phase14 snapshot, the extractor produces 67 CODE edges in a deterministic, idempotent pass; introduces no new maxim violations specific to CODE; and required no modifications to the schema, the maxim engine, or any pre-existing extractor — the schema's slot vocabulary already named the relation; only the extractor that materialises it needed to be written. The PEER_CONTRADICTION slot also gains its first edge in phase14 (1 edge), marking the first explicit declaration of a known conflict between research artifacts. The same path closes the remaining two slots: SUCCESSOR will be populated when scheduled follow-up experiments are extracted from the natural-language successor: annotations currently buried in experimental entries; THERMO will be populated when thermodynamic measurements (entropy estimates, free-energy proxies) are coupled to their structural sources. Each absent slot names not a gap in the schema but a knowledge-management function that the research programme has not yet operationalized, and the schema's ability to name these absences precisely — together with the now-demonstrated extensibility — is itself evidence of its diagnostic utility.

7.2 Relation to Existing Ontological Frameworks

The GCT is not a competitor to established foundational ontologies such as BFO, DOLCE, or the OWL-based frameworks that have become the infrastructure of formal knowledge representation in domains from biomedicine to geoscience. It is a complementary layer that operates above the level at which those frameworks make their primary contributions. A BFO-compliant domain ontology provides the controlled vocabulary for entities and their universals — the stable type structure of a domain as it exists independent of any particular investigation. The GCT provides what foundational ontologies by design do not: a causal-dynamic schema for representing how entities produced by a scientific investigation relate to one another across time, scale, and epistemic confidence, where the edges themselves carry provenance,

confidence scores, and scale metadata that encode the investigative process rather than merely its outputs.

The natural integration architecture treats GCT node labels as references into a BFO-compliant namespace while the GCT's edge schema provides the relational infrastructure. In this arrangement, a GCT node typed as `experiment` could resolve to a BFO `planned process`, while a node typed as `concept_functional_use` could resolve to a DOLCE non-physical `endurant`; the GCT then contributes the `CAUSE`, `INSTANTIATION`, `PREDECESSOR`, and other typed edges that connect these BFO-typed entities across the investigation's temporal and causal structure. The relationship is analogous to that between PROV-O and domain ontologies: PROV-O provides a provenance vocabulary that tracks the processes by which information artifacts were produced and transformed, complementing rather than replacing the domain vocabularies that characterize what those artifacts contain. Future work should formalize the GCT-BFO alignment via an OWL ontology that maps each of the twelve slots to the nearest BFO relation where a defensible correspondence exists — `CAUSE` to causally upstream of, `INSTANTIATION` to `is_a` — and introduces new OWL object properties for slots that have no BFO analog, such as `PREHENSION`. A first-pass OWL serialization of this mapping ships alongside this paper as `gct.owl` and is parseable by `rdflib v6+`; see `publication/gct_owl_README.md` for the slot-by-slot mapping decisions and a worked example.

7.3 Limitations

Three principal limitations qualify the claims advanced in this paper. The first concerns the geometric derivation of the twelve-slot schema. The schema is derived from the combinatorial structure of the cuboctahedron — specifically, from its twelve vertices, each representing one relational slot — and the cuboctahedron is selected on the grounds that it is the Vector Equilibrium, the unique configuration in which all edge lengths equal the radius, producing a schema in which no relational type is geometrically privileged over any other. This is an argument from the specific theoretical commitments of the research programme in which the GCT was developed, not a derivation from first principles of knowledge representation theory. Alternative geometric bases would yield schemas of different cardinality: an icosahedral basis would produce thirty slots, almost certainly too fine-grained for most research corpora; a tetrahedral basis would produce four, almost certainly too coarse. The argument that twelve is the correct cardinality rests on the Vector Equilibrium's formal properties, but whether those properties constitute a knowledge-representation-theoretic justification or a felicitous structural analogy remains an open question that a single corpus evaluation cannot resolve.

The second limitation concerns the inferential scope of the Boethian maxim engine. The current implementation performs monotone deductive inference: if the premises of a maxim are satisfied, its conclusion is asserted without qualification, and no mechanism exists for retracting a conclusion should subsequent evidence undermine the premise. Many real-world epistemic relations in scientific research are irreducibly defeasible — an experiment is a *prima facie* cause of a claim, but the claim may be retracted if the experiment is found to be confounded. Extending the maxim system to handle defeasible inference would require integrating a non-monotone reasoning layer, such as a default logic or an argumentation framework, that can represent the conditional and revisable character of scientific evidence.

The third limitation is the single-corpus evaluation design. The structural arguments for the GCT's domain independence rest on structural analogies rather than on empirical evaluation across multiple domains. A schema designed around the relational needs of a computational neuroscience programme might instantiate different slot distributions, surface different unoccupied slots as diagnostically informative, and require different manual edge patches in a materials science or linguistics programme. Formal evaluation of the GCT's generality requires replication in at least two additional domains with substantially

different ontological commitments, a task that constitutes the most important direction for future work.

8. Conclusion

We have presented the Tensegrity Knowledge Graph (GCT), a formally grounded architecture for the representation of scientific knowledge artifacts and their causal, taxonomic, evidential, and temporal relations. The system is defined as a typed directed multigraph $G = (N, E, \tau_N, \tau_E, \sigma, \lambda)$ equipped with a 12-slot semantic schema Σ derived from the geometry of Fuller's cuboctahedron, a five-level holonic scale hierarchy derived from Koestler's concept of the holon, and an inference engine whose twelve maxims are formalised adaptations of Boethius's topical reasoning. Seven classical philosophical traditions — Fuller, Koestler, Aristotle, Whitehead, Wittgenstein, Porphyry, and Boethius — are operationalised as distinct computational components, each responsible for a separable representational function. Soundness of the maxim engine with respect to holonic scale constraints has been established formally, and Aristotelian completeness has been defined as a computable property measurable on individual nodes. Empirical evaluation over 1,660 nodes and 2,120 typed edges from an active research programme confirmed the consistency and productivity of the inference layer, surfaced an 84.0% rate of formal incompleteness in structural nodes, and established that god nodes emerge from graph structure without manual designation. An isomorphic instantiation in biomedical research management established domain independence: the architecture is not a domain-specific encoding but a general schema for intentional artifact networks.

The GCT demonstrates that classical philosophical traditions which predate computation by centuries carry direct computational consequences when translated faithfully into typed graph schemas. Aristotle's four causes are not merely a classificatory scheme; when encoded as distinct edge types, they license categorically different inference rules and make the difference between a system that can distinguish design-compliance from causal attribution and one that cannot. Boethius's topical maxims are not merely rhetorical heuristics; when formalised as graph-theoretic constraints over typed edges and holonic scales, they constitute a sound inference engine that operates without an external reasoner. Wittgenstein's insistence that meaning is constituted by use, not by necessary-and-sufficient conditions, is not merely a philosophical position; when operationalised as dual similarity metrics over co-occurrence and neighbour-set structure, it provides coverage for the full range of experimental nodes in a live research corpus. The IVM cuboctahedron provides the 12-slot schema with a principled non-arbitrariness: because every vertex of the cuboctahedron is equidistant from the centre and every slot subtends an equal solid angle, no relational dimension in Σ is geometrically privileged over any other. These traditions converge not because they were forced together but because they all address the same structural problem — how to represent intentional artifacts in a way that is simultaneously causal, taxonomic, temporal, and inference-enabling — and the geometry of that problem is what the cuboctahedron encodes.

Three directions govern the near-term research agenda. First, formal alignment of GCT with BFO and serialisation to OWL would position the architecture for interoperability with the OBO Foundry ecosystem; the primary technical challenge is mapping the holonic scale hierarchy to BFO's universal-particular distinction without collapsing the scale-indexed semantics that GCT's inference engine depends on. Second, extending the Boethian maxim engine to support non-monotone defeasible reasoning — in which some Aristotelian inferences are *prima facie* warranted but rebuttable by later evidence — would bring the system into alignment with the actual epistemic dynamics of scientific research. Third, a controlled user study measuring the effect of GCT-guided knowledge management on the detection rate of contradictions, undocumented formal causes, and temporally decayed claims, compared against unstructured alternatives, would provide the external validity evidence needed to recommend the architecture for adoption in research management practice.

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Data and Code Availability

The complete reproducibility package — including the frozen `tensegrity_graph_phase9_complete.json` snapshot (SHA-256 `dc7bf9d5...d1953739`), the companion `tensegrity_graph_phase14_complete.json` (§7.1), the `tensegrity_graph_biomedical.json` second-corpus snapshot (§6), the OWL serialisation `gct.owl` (240 triples, OWL 2 DL profile), and all reproducibility and figure-generation scripts — is available at <https://github.com/expelius/TensegrityProject> under the `gct-paper-v1` release tag. A Zenodo DOI will be minted upon paper acceptance. Reviewers can verify every numerical claim in §5 by running `python publication/compute_paper_numbers.py` from the package root; the script reports the canonical pre-computed metrics directly from the snapshot SHA.

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